



Intel P35 Motherboards: ASUS P5K Deluxe / Foxconn P35A / Gigabyte GA-P35-DQ6 / MSI P35 Platinum

Bearlake 4 under the scanner

Intel's new P35 chipset (code-named Bearlake) marks the launch of yet another milestone for Desktop computing, namely DDR3 memory. Though the chipset supports DDR3, in the initial phase, motherboards will be offered with DDR2 memory to streamline the transition to DDR3. Memory speed apart, the new chipset ushers in support for the new, yet-to-be-launched Penryn processors (45nm at 1333 MHz FSB).

In this round-up, we have four motherboards: the ASUS P5K Deluxe, MSI Platinum, Gigabyte GA-P35-DQ6, and the Foxconn P35A. The former three are geared towards enthusiasts (read: bells and whistles attached).

Features

All four motherboards use the ATX form factor, and hence demand a full ATX cabinet. And all of them have taken to silent cooling. The MSI Platinum has oodles of copper pipes running around the Northbridge, with fins to

increase surface area. ASUS has its own implementation; Gigabyte goes a step forward and includes a back plate to cool the PCB. These substantial coolers will impede installation of a larger heatsink-fan combo for the processor.

The same pipes run towards the Southbridge, terminating into either a copper heatsink with fins (Gigabyte, MSI) or without them (ASUS). The Foxconn motherboard hasn't gone with silent cooling—this is their P35 for the masses.

The placement of the 24-pin ATX connector is spot-on on the ASUS and the Foxconn boards, but the dense heat pipes on the Gigabyte and the MSI hinder easy installation of the ATX connector. On all the boards, extra power is available via a four-pin Molex connector.

The memory slots on the ASUS and Foxconn don't interfere with the graphics card. Sadly, Gigabyte and MSI miss the mark by a few inches; the Corsair Dominator (as will any memory with a bulky heat spreader) comes dangerously close to the graphics card's PCB. And with the MSI and ASUS, a long

graphics card will run over the SATA ports.

Gigabyte's DQ6's Front Panel Audio gets trapped in between the back panel I/O ports and the thermal pipe solution, so Gigabyte implements a pin-raiser for easy access to it. The USB extension headers are placed at the bottom of all the boards, with ASUS and Foxconn using plastic brackets for trouble-free installation.

Gigabyte implements three PCIe x1 slots, whereas the ASUS P5K Deluxe and MSI P35 Platinum have two. The Foxconn provides just one. ASUS and Foxconn implement three PCI slots; MSI and Gigabyte, two each.

The ASUS lacks a PS2 port and alongwith P35 Platinum both of them have six USB, one FireWire, and two eSATA ports. The Foxconn P35A and Gigabyte GA-P35-DQ6 both provide four USB ports. The Foxconn also has an eSATA port.

All the boards have 8-channel audio; the ASUS P5K Deluxe and the Gigabyte GA-P35-DQ6 have multi-streamed audio, a technology that allows separate signals to be sent to the rear speaker connectors

and headphones, separating background audio streams from communications applications.

Power And The BIOS

Gigabyte has opted for a 12-phase power design—supposedly good for overclocking. ASUS offers 8-phase power, which should be good enough for driving the CPU to crazy overlocks. The Foxconn P35A and MSI P35 Platinum use a simpler, 4-phase power—no problems for stock systems, but could limit the CPU's overclocking potential.

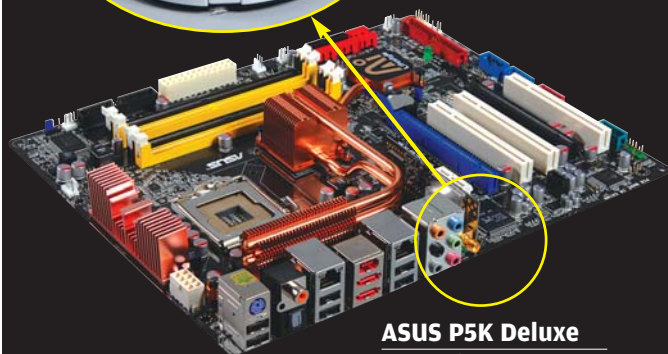
Though 1066 MHz DDR2 RAM is supported on all boards, the Foxconn and MSI required adjustment to run the Corsair Dominator at 1066 MHz.

The ASUS has a feature-rich BIOS with controls for the North / Southbridge voltage, and also for the CPU Damper, Transaction Booster, and Clock Over-Charging Mode.

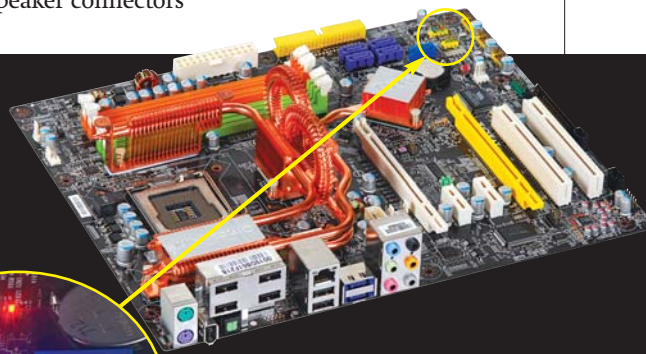
The Gigabyte C.I.A 2 has five preset overclocking profiles, whereas the ASUS provides two user-configurable overclocking profiles. All the boards come with utilities for



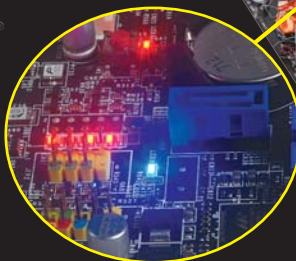
← Wi-Fi antenna with magnetic base



ASUS P5K Deluxe
Rs 14,750



MSI P35 Platinum
Rs 13,000



← 8 Diagnostic LED's: Easy Diagnosis of boot-up failures.